Model parțial
-ALGAED-

(4) a) Fie bazele $B_1, B_2 \subseteq \mathbb{R}^3$; $B_1 = \{(1,3,3); (1,2,4); (2,5,8)\}$ și $B_2 = \{(1,3,4); (1,2,5); (1,3,5)\}$. Fie $u \in \mathbb{R}^3$. Determinați coordonate lui u în baza B_1, dacă în baza B_2 u are coord: $3, -3, 2$.

$$\Rightarrow u = 3(1,3,4) - 3(1,2,5) + 2(1,3,5) =$$
$$= (3,9,12) - (3,6,15) + (2,6,10) =$$
$$= (2,9,7)$$

Fie $\alpha, \beta, \gamma \in \mathbb{R}$ a.î. $u = \alpha(1,3,3) + \beta(1,2,4) + \gamma(2,5,8)$

$$(2,9,7) = (\alpha+\beta+2\gamma; 3\alpha+2\beta+5\gamma; 3\alpha+4\beta+8\gamma)$$

$$\Rightarrow \begin{cases} \alpha+\beta+2\gamma = 2 \\ 3\alpha+2\beta+5\gamma = 9 \\ 3\alpha+4\beta+8\gamma = 7 \end{cases} \Rightarrow 3\alpha+3\beta+6\gamma = 6 \;\Big|\; \ominus \Rightarrow \beta+\gamma = -3 \;|\cdot(2)$$

$$\ominus \quad 2\beta+3\gamma = -2$$
$$2\beta+2\gamma = -6$$
$$/ \quad \gamma = 4 \Rightarrow \beta+4 = -3 \Rightarrow \beta = -7$$

$$\Rightarrow \alpha - 7 + 8 = 2 \Rightarrow \alpha = 1$$

$\Rightarrow$ coord lui u în baza B_1 sunt $1, -7, 4$

b) Fie $f: \mathbb{R}^3 \to \mathbb{R}^3$, cu $\dim_{\mathbb{R}} \operatorname{Ker} f = 2$. Avem $a, b, c, d \in \mathbb{R}$ astfel încât: $f(1,3,5) = (2,4,6)$; $f(2,3,3) = (5,a,b)$ și $f(3,4,5) = (6,c,d)$. Determinați a, b, c, d.

Dim Th rang-defect: $\dim_{\mathbb{R}} \operatorname{Im} f = \dim_{\mathbb{R}} \mathbb{R}^3 - \dim_{\mathbb{R}} \operatorname{Ker} f =$
$= 3 - 2 = 1 \;\Big|\; \Rightarrow$

Știm că $(2,4,6) \in \operatorname{Im} f \Rightarrow 2\cdot(1,2,3) \in \operatorname{Im} f \;\Big|$

$\Rightarrow \operatorname{Im} f = \operatorname{Sp}\{(1,2,3)\} = \{\alpha\cdot(1,2,3) \mid \alpha \in \mathbb{R}\}$

Dacă $(5,a,b) \in \operatorname{Im} f \Rightarrow (5,a,b) = \beta(1,2,3) \Rightarrow \beta = 5 \Rightarrow$
$\Rightarrow a = 10;\ b = 15.$

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Dacă $(6,c,d) \in \operatorname{Im} f \Rightarrow (6,c,d) = f(1,2,3) \Rightarrow \gamma = 6 \Rightarrow$ (1)(2)

$\Rightarrow c = 12 \; ; \; d = 18.$

3. Fie $f: \mathcal{M}_2(\mathbb{R}) \to \mathcal{M}_2(\mathbb{R})$, $f\begin{pmatrix} x_1 & x_2 \\ x_3 & x_4 \end{pmatrix} = \begin{pmatrix} x_1+3x_2+x_3-2x_4 & -2x_1-5x_2-2x_3+5x_4 \\ 3x_1+10x_2+3x_3-5x_4 & x_2+x_4 \end{pmatrix}$. Determinați câte o bază pt subspațiile $\operatorname{Ker} f + \operatorname{Im} f$, $\operatorname{Ker} f \cap \operatorname{Im} f$.

$$\operatorname{Ker} f = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathcal{M}_2(\mathbb{R}) \;\middle|\; f\begin{pmatrix} a & b \\ c & d \end{pmatrix} = O_{\mathcal{M}_2(\mathbb{R})} \right\}$$

$$\Rightarrow \begin{pmatrix} x_1+3x_2+x_3-2x_4 & -2x_1-5x_2-2x_3+5x_4 \\ 3x_1+10x_2+3x_3-5x_4 & x_2+x_4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} x_1+3x_2+x_3-2x_4=0 \\ -2x_1-5x_2-2x_3+5x_4=0 \\ 3x_1+10x_2+3x_3-5x_4=0 \\ x_2+x_4=0 \Rightarrow x_4=-x_2 \end{cases}$$

$$\Leftrightarrow \begin{cases} x_1+5x_2+x_3=0 \Rightarrow x_2 = \frac{-1}{5}(x_1+x_3) \\ -2x_1-10x_2-2x_3=0 \\ 3x_1+15x_2+3x_3=0 \end{cases}$$

$$\Rightarrow \operatorname{Ker} f = \left\{ \begin{pmatrix} x_1 & -\frac{1}{5}(x_1+x_3) \\ x_3 & \frac{1}{5}(x_1+x_3) \end{pmatrix} \;\middle|\; x_1, x_3 \in \mathbb{R} \right\} =$$

$$= \left\{ \frac{1}{5}\begin{pmatrix} 5x_1 & -x_1-x_3 \\ 5x_3 & x_1+x_3 \end{pmatrix} \;\middle|\; x_1, x_3 \in \mathbb{R} \right\} =$$

$$= \left\{ \frac{1}{5}x_1 \underbrace{\begin{pmatrix} 5 & -1 \\ 0 & 1 \end{pmatrix}}_{v_1} + \frac{1}{5}x_3 \underbrace{\begin{pmatrix} 0 & -1 \\ 5 & 1 \end{pmatrix}}_{v_2} \;\middle|\; x_1, x_3 \in \mathbb{R} \right\} =$$

$$= \operatorname{Sp}\{v_1, v_2\} \Rightarrow \{v_1, v_2\} \text{ SG pt } \operatorname{Ker} f. \; (1)$$

Fie $\alpha, \beta \in \mathbb{R}$ a.ș. $\alpha v_1 + \beta v_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow \alpha\begin{pmatrix} 5 & -1 \\ 0 & 1 \end{pmatrix} + \beta\begin{pmatrix} 0 & -1 \\ 5 & 1 \end{pmatrix} =$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow \begin{pmatrix} 5\alpha & -\alpha-\beta \\ 5\beta & \alpha+\beta \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \Leftrightarrow \alpha = \beta = 0 \Rightarrow$$

$\Rightarrow \{v_1, v_2\}$ SLI (2)

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(1)(2) $\Rightarrow$ $\{v_1, v_2\}$ o bază pt $\ker f$ și $\dim_{\mathbb{R}} \ker f = 2$

Dim Th rang-defect $\Rightarrow \dim_{\mathbb{R}} \operatorname{Im} f = \dim_{\mathbb{R}} M_2(\mathbb{R}) - \dim_{\mathbb{R}} \ker f =$

$= 4 - 2 = 2$

Fie B baza canonică a lui $M_2(\mathbb{R}) \Rightarrow$

$\Rightarrow B = \{e_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}; e_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; e_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; e_4 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\}$

$\Rightarrow f(e_1) = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}; f(e_2) = \begin{pmatrix} 3 & -5 \\ 10 & 1 \end{pmatrix}; f(e_3) = \begin{pmatrix} 1 & -2 \\ 3 & 0 \end{pmatrix}$

$f(e_4) = \begin{pmatrix} -2 & 5 \\ -5 & 1 \end{pmatrix}$

Scriem A matricea relativă la baza B a lui f.

$$A = \begin{pmatrix} 1 & 3 & 1 & -2 \\ -2 & -5 & -2 & 5 \\ 3 & 10 & 3 & -5 \\ 0 & 1 & 0 & 1 \end{pmatrix} \xrightarrow[L_3 \to L_3 - 3L_1]{L_2 \to L_2 + 2L_1} \begin{pmatrix} 1 & 3 & 1 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix} \xrightarrow[L_4 \to L_4 - L_2]{L_3 \to L_3 - L_2}$$

$f(e_1)\ f(e_2)\ f(e_3)\ f(e_4)$

$$\longrightarrow \begin{pmatrix} \textcircled{1} & 3 & 1 & -2 \\ 0 & \textcircled{1} & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Rightarrow \{f(e_1); f(e_2)\} \text{ bază pt } \operatorname{Im} f.$$

$f(e_1)\ f(e_2)\ f(e_3)\ f(e_4)$

Luăm matricea A' care are pe coloane coordonatele vectorilor $v_1; v_2; f(e_1); f(e_2)$ în baza B

$$A' = \begin{pmatrix} 5 & 0 & 1 & -2 \\ -1 & -1 & -2 & 5 \\ 0 & 5 & 3 & -5 \\ 1 & 1 & 0 & 1 \end{pmatrix} \xrightarrow{L_1 \leftrightarrow L_4} \begin{pmatrix} 1 & 1 & 0 & 1 \\ -1 & -1 & -2 & 5 \\ 0 & 5 & 3 & -5 \\ 5 & 0 & 1 & -2 \end{pmatrix} \xrightarrow[L_4 \to L_4 - 5L_1]{L_2 \to L_2 + L_1}$$

$v_1\ v_2\ f(e_1)\ f(e_2)$

$$\longrightarrow \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & -2 & 6 \\ 0 & 5 & 3 & -5 \\ 0 & -5 & 1 & -7 \end{pmatrix} \xrightarrow{L_3 \to L_3 + L_4} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & -2 & 6 \\ 0 & 0 & 4 & 2 \\ 0 & -5 & 1 & 7 \end{pmatrix} \xrightarrow{L_2 \leftrightarrow L_4} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & -5 & 1 & 7 \\ 0 & 0 & 4 & 2 \\ 0 & 0 & -2 & 6 \end{pmatrix}$$

$$\xrightarrow{L_3 \to L_3 + 2L_4} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & -5 & 1 & 7 \\ 0 & 0 & 0 & 14 \\ 0 & 0 & -2 & 6 \end{pmatrix} \xrightarrow[L_3 \to \frac{1}{14} L_3 \\ L_4 \to \frac{-1}{2} L_4]{L_2 \to \frac{-1}{5} L_2} \begin{pmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & -1/5 & -7/5 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -3 \end{pmatrix} \xrightarrow{L_3 \leftrightarrow L_4}$$

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$$\rightarrow \begin{pmatrix} \textcircled{1} & 1 & 0 & 1 \\ 0 & \textcircled{1} & -1/5 & -7/5 \\ 0 & 0 & \textcircled{1} & -3 \\ 0 & 0 & 0 & \textcircled{1} \end{pmatrix} \Rightarrow \{v_1, v_2, f(e_1); f(e_2)\}$$ o bază pt $\ker f + \operatorname{Im} f$

$v_1\ v_2\ f(e_1)\ f(e_2)$

$\Rightarrow \dim_{\mathbb{R}}(\ker f + \operatorname{Im} f) = 4$

Dim Th Grassmann $\Rightarrow \dim_{\mathbb{R}}(\ker f \cap \operatorname{Im} f) = \dim_{\mathbb{R}} \ker f + \dim_{\mathbb{R}} \operatorname{Im} f - \dim_{\mathbb{R}}(\ker f + \operatorname{Im} f) = 2+2-4 = 0$

$\Rightarrow \ker f \cap \operatorname{Im} f = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow$

$\Rightarrow \{0_2\}$ o bază pt $\ker f + \operatorname{Im} f$.

1. Fie punctul $A(1,2,4)$ și dreapta $d: \frac{x+2}{2} = \frac{y-3}{-1} = \frac{z+3}{3}$. Determinați distanța de la A la d.

$d: \frac{x+2}{2} = \frac{y-3}{-1} = \frac{z+3}{3} \overset{\text{not}}{=} t \in \mathbb{R} \Rightarrow$

$$\Rightarrow \begin{cases} x = 2t-2 \\ y = -t+3 \\ z = 3t-3 \end{cases} ; t \in \mathbb{R}$$

Fie $A' = pr_d A$ și Q un plan a.î. $A \in Q$ și $d \perp Q \Rightarrow$

$\Rightarrow \vec{v_d} = \vec{N_Q} = 2\vec{i} - \vec{j} + 3\vec{k} \Rightarrow Q: 2(x-1) - (y-2) + 3(z-4) = 0 \Rightarrow$

$\Rightarrow Q: 2x - y + 3z - 12 = 0$ $\Big|$ $\rightarrow 2(2t-2) - (-t+3) + 3(3t-3) - 12 = 0$

Dar $d \cap Q = \{A_1\}$

$\Rightarrow 4t - 4 + t - 3 + 9t - 9 - 12 = 0 \Rightarrow 14t = 28 \Rightarrow t = 2$

$\Rightarrow A'(2,1,3)$

$\Rightarrow d(A,d) = \|\vec{AA'}\|$

$\vec{AA'} = (x_{A'} - x_A)\vec{i} + (y_{A'} - y_A)\vec{j} + (z_{A'} - z_A)\vec{k} =$

$= \vec{i} - \vec{j} - \vec{k}$

$\Rightarrow \|\vec{AA'}\| = \sqrt{3} \rightarrow d(A,d) = \sqrt{3}$

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Fie polinoamele $P_1(X) = 1+2X+3X^2+6X^3$; $P_2(X) = 5+6X+7X^2+2X^3$; $P_3(X) = 6+7X+aX^2+bX^3$. Determinați valorile parametrilor $a, b \in \mathbb{R}$ pentru care $\{P_1(X); P_2(X); P_3(X)\}$ să fie SLD.

Fie A matricea care are coord polinamelor $P_1(X)$; $P_2(X)$ și $P_3(X)$ din baza canonică a lui $\mathbb{R}_3[X]$ pe coloane.

$$A = \begin{pmatrix} 1 & 5 & 6 \\ 2 & 6 & 7 \\ 3 & 7 & a \\ 6 & 2 & b \end{pmatrix}$$
$P_1\ P_2\ P_3$

Dacă $\operatorname{rang} A \leq 2 \Rightarrow \{P_1, P_2, P_3\}$ SLD

$$A = \begin{pmatrix} 1 & 5 & 6 \\ 2 & 6 & 7 \\ 3 & 7 & a \\ 6 & 2 & b \end{pmatrix}$$

$$\Delta_2 = \begin{vmatrix} 1 & 5 \\ 2 & 6 \end{vmatrix} = 6-10 = -4 \neq 0 \Rightarrow \operatorname{rang} A = 2$$

$$\Delta_{3_1} = \begin{vmatrix} 1 & 5 & 6 \\ 2 & 6 & 7 \\ 3 & 7 & a \end{vmatrix} = 6a+84+105-(108+49+10a) = 6a+189-157-10a =$$
$$= -4a-32 = 0 \Leftrightarrow a = 8$$

$$\Delta_{3_2} = \begin{vmatrix} 2 & 6 & 7 \\ 3 & 7 & 8 \\ 6 & 2 & b \end{vmatrix} = 14b+42+288-(294+18b+32) =$$
$$= 14b+330-18b-326 =$$
$$= -4b+4 = 0 \Leftrightarrow b = 1$$

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